pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

Five Questions Without Retrieval

WEEK 1 · DAY 1

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: five questions, no document

The document and the questions

Document	Boyd Gaming, 10-K, fiscal 2025
Print	the course's PDF, 113 pages
Questions	75 in the set; 5 used today
Keys	written first, each a quote
Answers	saved; replayed, no key

Today

1. What RAG is; the course's arc
2. Where Course 7 stopped
3. The key, the mode, the transport
4. Five questions, keys first
5. Closed book, graded
6. Prompt only, graded
7. Ten answers: a table and charts

No document goes in any request today. That is the lesson.

The rules for retrieval

Cite or say so	every sentence cites a chunk, or NOT IN CORPUS
No market views	no opinion, grade, ordering, or view of the future
Issuers as authors	Boyd Gaming wrote the 10-K; nothing more
Text, not a view	what a model returns is text, read from saved files

What a generated answer is called

Allowed names

the answer in the saved run

the model returned

Never called

the model's view

advice

What RAG is, in three sentences

1. Find the passages of a document that bear on a question.
2. Hand only those to a model, each labelled with where it came from.
3. Ask for an answer from them, citing each label, or NOT IN CORPUS.

Step 1-2	Step 3	Step 4	Step 5	Step 6
No document	**The pipeline**	**Quality, tuning**	**Evaluation, upkeep**	**Findings as counts**
day 1	days 2 to 5	days 6 to 10	days 11 to 14	5, 10, 15

Every part is checkable: what was found, what was sent, what each sentence stands on.

Where Course 7 stopped

Course 7, day 14	What Course 10 adds, and the day	
one indenture, 454 chunks	pages and items from a PDF	**day 2**
180 words, by section	a recursive splitter	**day 3**
MiniLM, a NumPy product	an index, measured	**day 4**
labels like [Section 4.12]	BM25, hybrid, a reranker	**day 6, 7**
a deterministic judge	a tuning grid; several documents	**day 8, 9**
grounded is not right	citations checked; judges checked	**day 10, 12**

None of these pieces is needed today: no request carries a document.

The key, the mode, and the transport

rag_client: Course 8's transport

key = SHA-256 of method, path, body

never a header, never your key

replay: the saved response, or LookupError

live: sent, then saved under its key

Before any key

```
git check-ignore -v .env cache/
```

```
.gitignore:4:.env   .env
.gitignore:5:cache/   cache/
```

```
OpenAI key loaded: False. Mode: replay.
```

15 saved exchanges copied for day 1

```
One letter changed: LookupError: no saved exchange for request fef58c12c344 in day01
```

Five questions, keys first

Question, key (the key's page of the course's PDF)	
arc01	coupon and maturity of each senior notes series (p. 39, number key) `4.750% Senior Notes due 2031, maturing June 15, 2031; 4.750% Senior Notes due 2027, maturing December 1, 2027`
arc02	outstanding under the credit facility (p. 37, number key) `$160.7 million`
arc03	maturity of the revolving credit facility (p. 75, text key) `March 2, 2027`
arc04	the financial covenants and their levels (p. 38, number key) `A minimum consolidated interest coverage ratio of 2.50 to 1.00 and a maximum Consolidated Total Net Leverage ...`
arc05	variable-rate share; a 100 bp change (p. 48, number key) `About 7.8% variable; a 100-basis-point change moves annual interest costs by about $1.6 million`

8 of 8 key quotes found on their stated pages of the PDF.

How a closed-book answer works

The problem	A question about one filing's figures as of a date, put to a model not given the filing
The request	Two messages: Course 7's system prompt and the question; temperature 0, seed 42, 300 tokens
What returns	Text from training in the shape the prompt asks for: quotes and a section number, no passage behind them
You control	The model, the temperature, the seed, and max_tokens; the system prompt stays fixed
What comes out	ask's dict: content, the model as returned, the token counts; nothing in it can be cited

Closed book

Sent

system: Course 7's SYSTEM_PROMPT

user: the question

no document, no passage

arc02 in the saved run; key: $160.7 million

```
"Outstanding under the credit facility at December 31, 2025 was $1,200
million." (Section 3)
```

0 of 5

matched their keys

4 of 5

cite a section never sent

0 of 5

cite a page

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 5 requests, 697 prompt and 211 completion tokens; saved run recorded 2026-10-12; replayed (no key loaded).

```
chunker=none, embedder=none, index=none, retrieve=none k=0, rerank=none, expand=0, model=gpt-4.1-mini, temperature=0, top_p=1, max_tokens=300
```

How the grader works

The problem	Hundreds of answers marked right or wrong the same way every time, with no reader judging each
Number key	Pulled into its numbers, commas removed; a match only when every one is in the answer whole: 4.75 is not 4.750
Text key	Lower case, spaces squeezed, searched for whole; NOT IN CORPUS and REFUSED labelled before any key is read
You control	The key and its kind, written from the page first; the grader has no setting
What comes out	One of four labels: match, no match, not in corpus, refused; 1.8 draws the parts each answer held

What a closed-book answer is

3 of 5

state a year

0 of 5

cite a page

4 of 5

cite a section

Text from training: no document was sent

No source: nothing shows which filing a figure came from.

No date unless it writes one: a 10-K figure is as of a date, in one filing.

A matching number, closed book, is a coincidence of digits until a source is shown.

Prompt engineering only

PROMPT_ONLY_RULE, a second system message (still no document)

```
Answer in one sentence. State each figure with its unit and the date it is as of, as you have it. If you do not have the figure for the date asked, reply exactly NOT IN CORPUS.
```

0 of 5

matched their keys

0 of 5

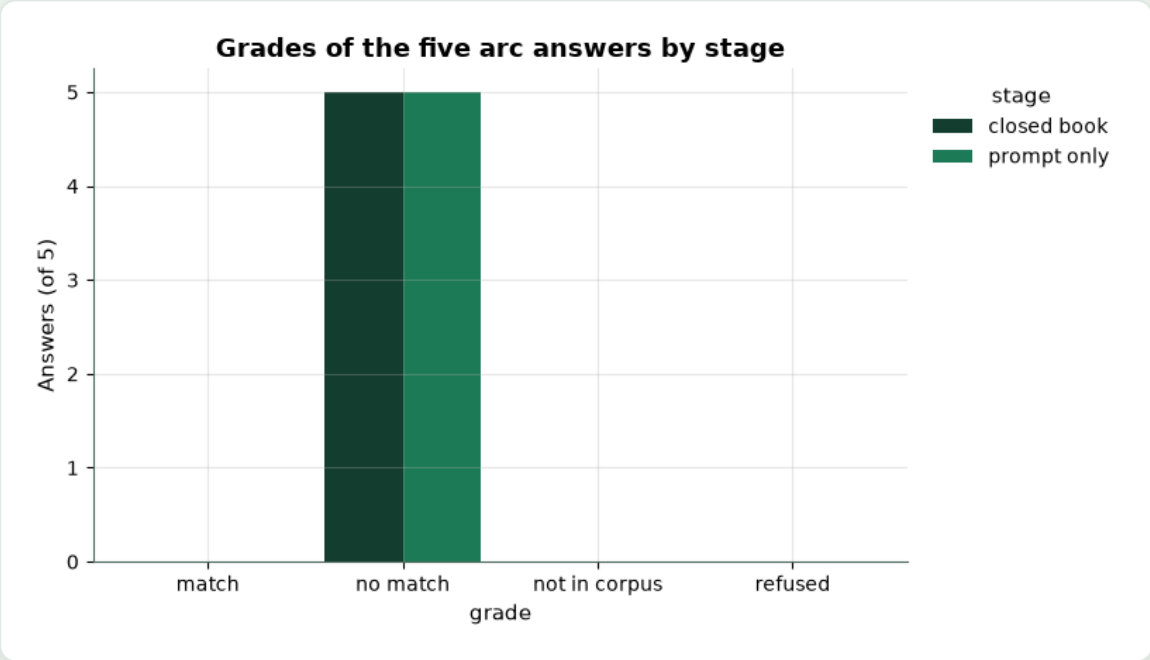
said NOT IN CORPUS

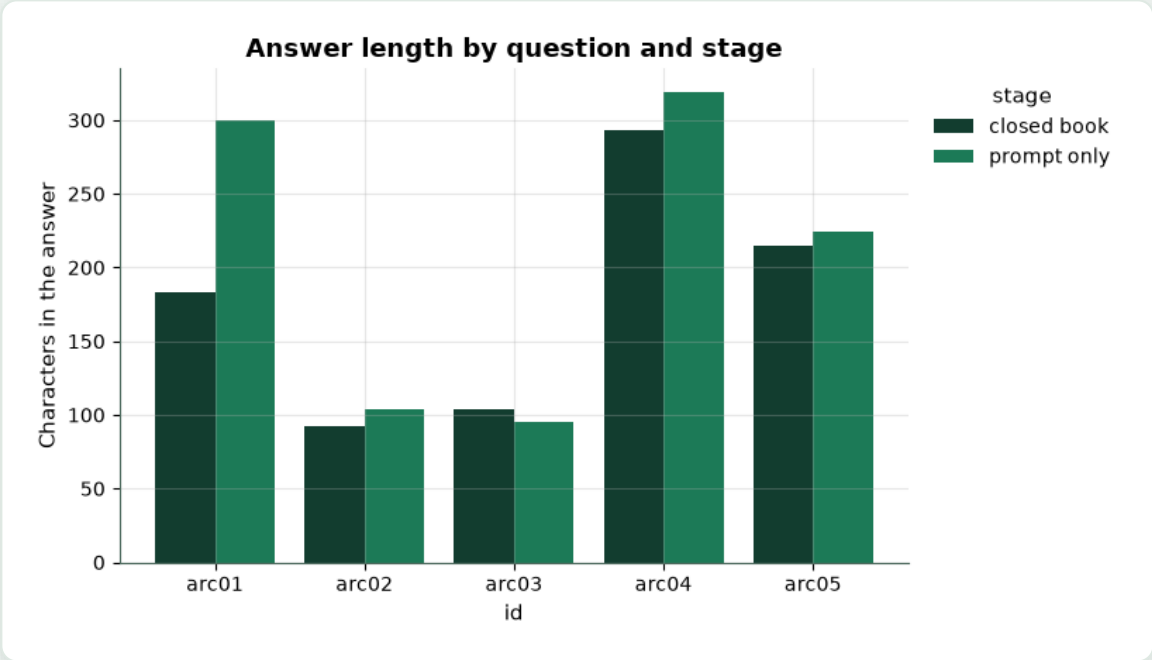
0 of 10

answers citing a page

A prompt can ask for a date and for NOT IN CORPUS. It cannot supply the document.

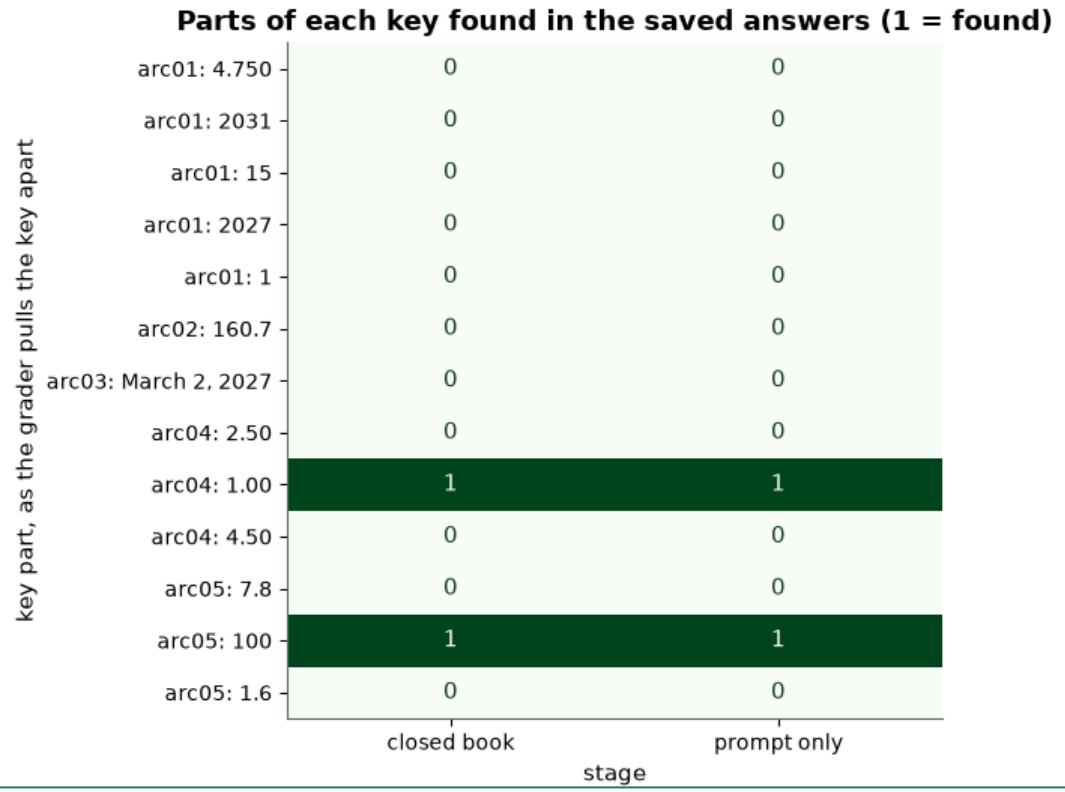
Ten answers against their keys





Closed book 0 of 5, prompt only 0 of 5 matched: without the document, every answer missed.

The grader, drawn



How to read it

one row per part of a key, as the grader cuts it

one column per stage; 1 = in the saved answer

a key matches only when all its rows are 1

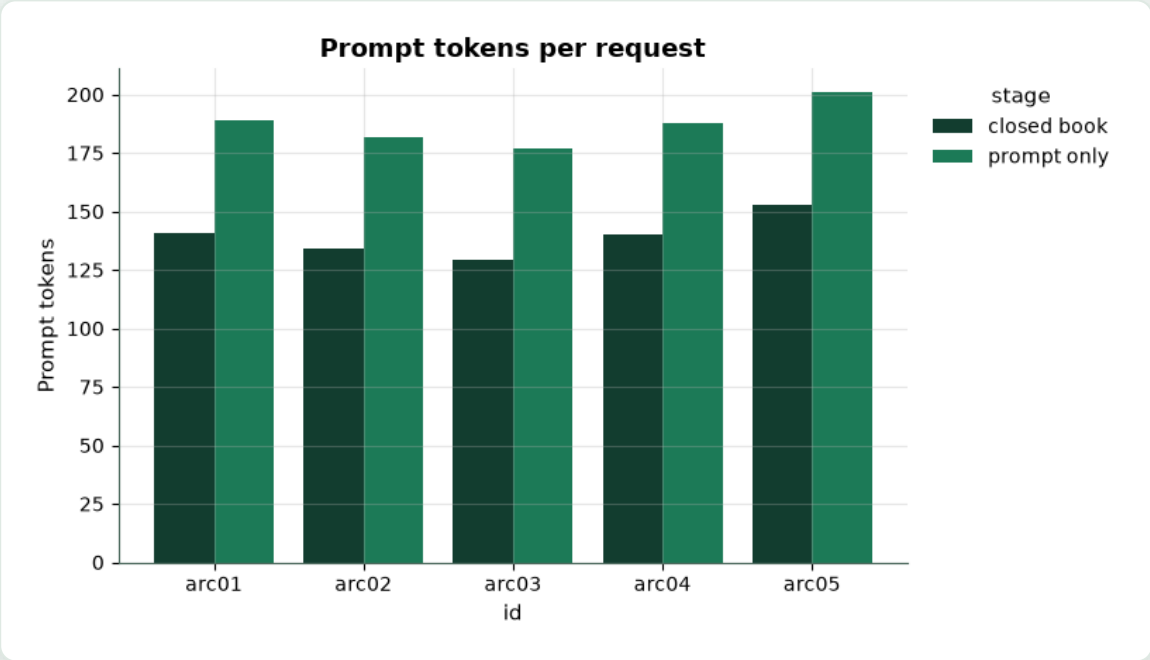
found: 2 of 13 closed book, 2 of 13 prompt only

1.00: a ratio's right side; 100: the question's bp

arc01 prompt only writes 4.75%: a 0 for 4.750

No key has every part in either column: the grades chart, row by row, and the whole-number rule where it bites.

Tokens and cost per request





Ten requests: 1,634 prompt and 501 completion tokens, US$0.001455 at the list price.

Prices: US$0.40 and US$1.60 per million tokens, developers.openai.com/api/docs/pricing, read 12 October 2026

Neither stage can cite a page; Excel and Git

0 of 10 cite a page; 9 of 10 cite a section of a document never sent.

Excel side by side (tested in Excel)

`=ISNUMBER(SEARCH(D4,E4))  (a text key)`	agreed on 2 of 2
`=AND(ISNUMBER(SEARCH("7.8",E6)),ISNUMBER(SEARCH("100",E6)),ISNUMBER(SEARCH("1.6",E6)))`	agreed on 8 of 8
`=COUNTIFS(B2:B11,"closed book",F2:F11,TRUE)`	0 closed book
`=ISNUMBER(SEARCH("4.75",E13))  (answer 4.750%)`	TRUE

One commit: five earlier modules beside ragkit, 10 tests. SEARCH finds 4.75 inside 4.750; the grader does not.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the last line returns match whenever both have a number: 5.250% passes for 4.750%.

Code that runs is not the same as code that is right.

What you can do now

- Say what RAG is, in three sentences
- Make the client with rag_client; key in .env, out of Git
- Write keys first, each a quote found on its page
- Ask closed book and prompt only; grade strictly
- Show a comparison as a styled table and a chart

Week 1, Day 2 A filing as pages and items